

Multilingual Video Dubbing System

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Problem Statement

Most videos on the internet exist in only one spoken language. A viewer who does not understand that language cannot fully use the video, even with subtitles, because subtitles still require constant reading, and many viewers skip or ignore them.

Making a proper dubbed version of a video today needs several separate steps, usually carried out by different people: writing down what is said, translating it, recording a new voice-over, timing the new audio so it matches the video, and re-editing the video with the new track. Each step depends on a different tool and a different skill. If the timing does not match, the final video feels unnatural, no matter how accurate the translation is.

Because of this, most independent creators, teachers, and small businesses simply do not translate their videos, and their content stays limited to the audience that already understands the original language.

Proposed Solution

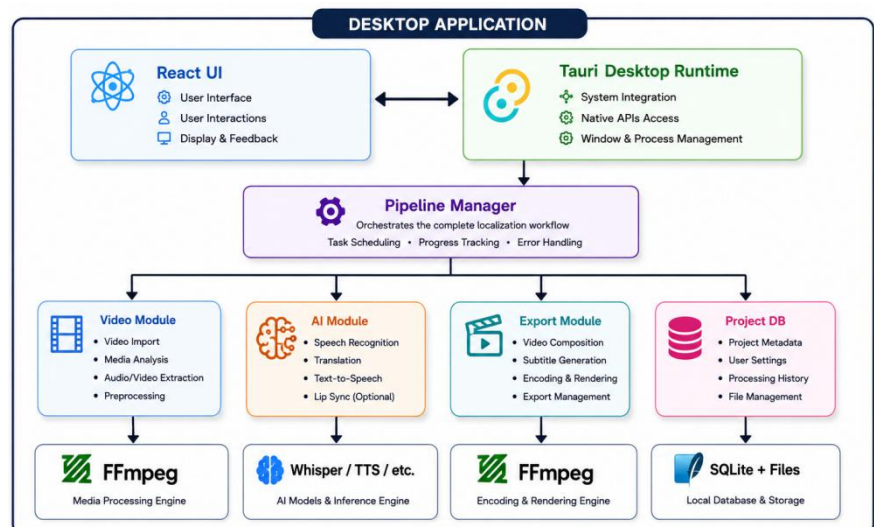
This project is a desktop application that takes a video in one language and produces a version in another language, complete with a new spoken voice track and matching subtitles. The user selects a video and a target language; the system carries out the rest of the process on its own device, through the following stages:

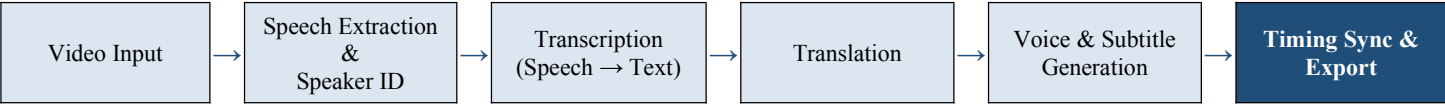
- Extract the spoken audio from the video.
- Identify how many people are speaking and separate their voices.
- Convert each speaker's audio into text, in the original language.
- Translate that text into the target language.
- Generate a new voice track in the target language, matched in timing to the original speech.
- Automatically generate subtitles for the translated speech.
- Combine the new audio, subtitles, and original video, and export the final file.

Everything runs on the user's own computer. No video or audio is uploaded to an external server, and **no paid subscription or API key is required for the core system to work**. The application is also built in separate modules, so a new language, a better AI model, or an extra processing feature can be added later without rebuilding the whole system.

System Workflow

The Pipeline Manager passes the video through each stage in order. Speech extraction and speaker identification happen together, so that later stages can generate one consistent voice per speaker instead of a single blended voice for the whole video. Timing sync runs last, adjusting the generated speech and subtitles so they line up with the original video before the final file is exported.





Key Technologies

The system is designed to run on a wide range of computers. A more powerful machine unlocks higher-quality AI models for each stage, while an ordinary laptop still runs the complete pipeline using lighter versions of the same models:

Component	Optimized for Low-End Hardware	For High-Computation Devices
Desktop Shell	Tauri	Tauri
User Interface	React + Tailwind CSS	React + Tailwind CSS
Core Language	Python	Python
Inter-Process Communication	Tauri Commands + Local API/IPC	Tauri Commands + Local API/IPC
Project Database	SQLite	SQLite
Configuration Storage	JSON	SQLite + JSON
Video Processing	FFmpeg	FFmpeg
Metadata Reading	FFprobe	FFprobe + MediaInfo
Speech Recognition	Faster-Whisper (Small / Base)	Whisper Large V3
Speaker Diarization	Skip / Basic Voice Detection	pyannote.audio
Language Detection	Whisper	fastText + Whisper
Translation	MarianMT / Distilled NLLB	NLLB-200 (3.3B)
Script Refinement	3B–7B instruction model	Larger instruction-tuned model
Text-to-Speech	Piper	XTTS-v2
Voice Cloning	Disabled	OpenVoice V2
Speech Timing Alignment	Rubber Band + FFmpeg	Advanced forced alignment
Lip Sync	Wav2Lip	MuseTalk
Subtitle Generation	pysrt	pysrt
Export	FFmpeg	FFmpeg
Model Management	Local model registry	Automatic download / update manager